



GREENROUTE CAMPUS

AI-powered sustainable campus transportation platform

Software-based Web Application · Flask · SQLite · Leaflet · Routing ·
Weather · Cloud AI

PRIMARY SDG ALIGNMENT

SDG 11 — Sustainable Cities and Communities

GreenRoute directly connects to sustainable mobility by helping users compare transportation choices and understand their time, cost and estimated emissions.



QR note: the included QR opens the presentation PDF through the local Flask server. On a phone, use the computer's LAN address instead of localhost when the app is shared over the same network.

United Nations SDG branding note

The SDG 11 icon shown here follows the official United Nations SDG 11 visual specification. The project is not endorsed or approved by the United Nations. For non-UN users, the UN guidance permits use of SDG icons subject to its branding guidelines; the UN emblem is not used here.

1. Project Overview

GreenRoute Campus is a software-based web application for sustainable campus mobility. It brings route planning, transport comparison, environmental impact estimates, trip logging, challenges, carpooling, dashboards and an AI Copilot into one platform.

2. Problem Statement

Students and staff can know where they need to go without having a clear view of the trade-offs between travel time, cost, convenience and environmental impact. Transport information is often fragmented across different tools. GreenRoute presents these factors together so the user can make an informed choice.

3. Objectives

- Compare multiple transportation modes using consistent calculations.
- Show estimated travel time, cost and CO₂ emissions.
- Provide route geometry through mapping/routing services when available.
- Add weather/AQI context to mobility decisions.
- Provide natural-language AI assistance.
- Record trips and turn individual actions into measurable campus data.

4. Main Modules

Smart Trip Planner · Interactive Map · Transport Comparison · AI Copilot · Impact Dashboard · Trip Ledger · Carpool Network · Challenges & Rewards · Leaderboard · Admin Command Center · Assumptions/Methodology page.

5. AI Component

The AI Copilot is designed as a genuine cloud-AI layer rather than a large if/else response table. GreenRoute first calculates feasible transport candidates on the server. The AI receives the user request plus those candidate metrics and contextual data, then selects and explains a candidate. The server validates the returned mode and keeps the displayed numerical metrics authoritative. An offline fallback keeps the demo usable if cloud AI is unavailable.

6. Technology Stack

Frontend: HTML, CSS, JavaScript. Backend: Python + Flask. Database: SQLite. Maps: Leaflet. Routing: OSRM with fallback calculations. Weather/AQI: external weather service with fallback. AI: OpenAI Responses API. Security: password hashing, CSRF protection, validation and rate limiting. PWA support is included for the responsive application shell.

7. SDG Alignment — SDG 11

United Nations Sustainable Development Goal 11 is “Sustainable Cities and Communities.” Its Target 11.2 specifically addresses access to safe, affordable, accessible and sustainable transport systems. GreenRoute’s focus on sustainable mobility, transport comparison and environmental awareness is therefore directly related to SDG 11. The project can also contribute to the broader environmental aims associated with SDG 13, but SDG 11 is the primary alignment.

8. How the System Works

PLAN → ACT → MEASURE → LEARN. The user enters a destination or selects a route. GreenRoute calculates route distance and transport candidates. The user can compare modes or ask the AI Copilot. The selected trip can be logged. Dashboards aggregate trip history and environmental metrics, creating a feedback loop for campus mobility analysis.

9. Expected Benefits

For students: clearer mobility choices and visible cost/time/environmental trade-offs. For the campus: structured mobility data, possible identification of repeated corridors and a basis for future interventions such as carpooling or shuttle planning. For sustainability education: a practical demonstration of how software and data can support environmental awareness.

10. Limitations & Assumptions

CO₂ values are estimates based on configured transport factors rather than direct vehicle measurements. Travel time depends on modeled speeds and routing availability. Weather/AQI data may be unavailable or delayed. AI output is explanatory and constrained by server-provided candidates; it should not be treated as a measurement system or guaranteed real-world forecast.

11. Future Scope

Live public transport feeds · campus shuttle tracking · real-time traffic · EV charging information · better campus-specific emission factors · mobile app packaging · institutional authentication · analytics for transport planning · multilingual AI assistance.

12. Suggested Demonstration Flow

Open the landing page → show SDG 11 alignment → log in → open Planner → select a route → compare modes → send context to AI Copilot → ask a natural-language question → show the AI explanation → log the trip → open Impact Dashboard → show challenges/leaderboard → finish with the Admin/analytics view.

Quick Viva / Teacher Explanation

What is it?	A software-based web application for sustainable campus transportation.
Why is it relevant?	It makes transport choices measurable by showing time, cost and estimated CO ₂ impact in one place.
Where is AI used?	The AI Copilot interprets natural-language requests and reasons over server-calculated transport candidates.
Which SDG?	SDG 11 — Sustainable Cities and Communities, especially the sustainable transport focus of Target 11.2.
What makes it practical?	It combines mapping, route calculations, environmental metrics, trip records and an interactive user interface rather than being only a static presentation.

Sources & attribution

United Nations, Sustainable Development Goal 11 — Sustainable Cities and Communities; United Nations SDG communications materials and SDG icon guidelines. See the official UN resources for the current icon and usage guidance.

UN guidance: <https://www.un.org/sustainabledevelopment/news/communications-material/> · SDG 11: <https://sdgs.un.org/goals/goal11>